

Enzyme formulation to improve cattle dung and wheat straw co-digestion in leach-bed reactor

Vitor Alves Lourenço^{1*}, Isabel Kimiko Sakamoto¹, Edson Luiz Silva², Maria Bernadete Amâncio Varesche¹, Renaud Escudié³, Hélène Carrère³

¹Department of Hydraulics and Sanitation, São Carlos School of Engineering, University of São Paulo (USP), São Carlos, SP, 13563120, Brazil.

*e-mail: vitor.a.lourenco@gmail.com; vitor.a.lourenco@usp.br

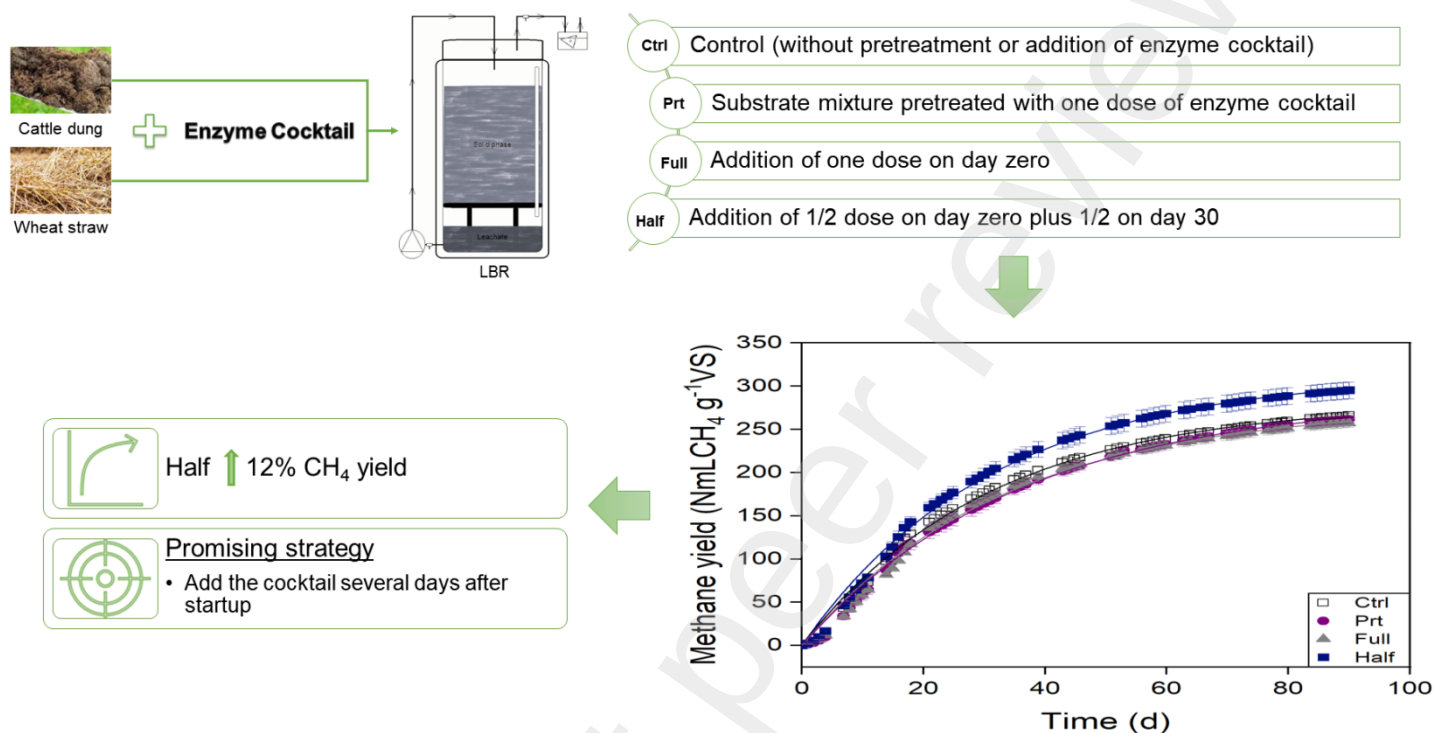
²Center of Exact Sciences and Technology, Department of Chemical Engineering, Federal University of São Carlos (UFSCar), São Carlos, SP, Brazil.

³INRAE, Montpellier University, Laboratory for Environmental Biotechnology (LBE), Narbonne, France.

*Corresponding author

1 Enzyme formulation to improve cattle dung and wheat straw 2 co-digestion in leach-bed reactor

3 4 Graphical abstract



6 7 Abstract

8 The objective of the study was to investigate the use of an enzyme formulation (5 g Kg⁻¹VS) in the co-digestion of cattle dung and wheat straw in leach-bed reactors. Three
9 strategies were tested: adding the formulation during the pre-treatment; at the start of the
10 batch; and at the start and during the process. Only the addition of half a dose on day zero
11 followed by half dose on day 30 increased methane production (12%) and favored
12 *Methanosarcina* predominance in solid (74%) and liquid phase (54%). The main bacteria
13 involved in hydrolysis, acidogenesis, and acetogenesis in the solid phase was
14 *Ruminofilibacter* (21%). The results highlight that in leach-bed reactor, it is crucial not to
15 overdose the enzyme at the start of the reactor to allow sufficient time for methanogen
16

growth. Therefore, the most effective use of enzymes is to add them at the startup and again after several days of operation.

Keywords: biogas, dry anaerobic digestion, solid-state, enzymatic pretreatment

Highlights:

- Pretreatment with 5 g enzymes Kg⁻¹VS or addition at day 0 resulted in VFA accumulation
- Full dose at day 0 had negative impact on hydrolysis and methanogenesis
- Increase in CH₄ yield (11.7%) with half doses on days 0 and 30
- There was no accumulation of VFA with the addition of half dose on day 30
- Half doses used on days 0 and 30 favored the growth of *Methanosarcina* (74.26%)

1. Introduction

Although agricultural development has significantly contributed to the global bioeconomy [1], food production results in a substantial amount of agricultural residues or waste. These residues primarily consist of lignocellulosic biomass (such as bagasse, bark, straw or stalk) and livestock waste (including manure, waste or animal carcasses). One of the most commonly used management practices for these organic wastes is their degradation to produce methane through digestion (or co-digestion) in anaerobic reactors [2]. In many regions, this biological process has already been successfully implemented and contributes to the production of clean energy [3].

These high-solid-content biomasses are particularly suited to dry anaerobic digestion processes. Among them, the leach-bed reactor (LBR), which operates in batch mode, is gaining attraction in rural contexts due to its economic and technological

advantages [4]. These advantages include low processing requirements for the digestate, which do not necessitate post-treatment [5]. In LBRs, the substrate is placed in a bed with the inoculum (solid digestate), while the liquid phase (leachate), stored at the bottom of the reactor or in a tank, is intermittently sprinkled over the solid phase and percolates through it.

However, the use of lignocellulosic waste presents challenges in achieving maximum biomass conversion to biogas due to its structure [6]. The cellulose, hemicellulose and lignin present in these residues have complex structures with chemical and physical recalcitrance that must be hydrolyzed/depolymerized to become accessible during the bioconversion process [1]. Therefore, it is recommended to pre-treat the waste or add agents that facilitate its degradation during the anaerobic digestion process.

Various types of enzymes have been used for the biological pretreatment of agricultural residues, yielding positive results across a wide range of substrates [1]. The use of enzymes for the management of agricultural waste is advantageous because their effects during pre-treatment extend to the main stages of hydrolysis and saccharification of biomass during the digestion process, according to the activity time of the enzymes used [7]. Furthermore, enzyme engineering concepts have been investigated at the molecular level to increase the catalytic efficiency of lignocellulolytic enzymes. The most viable and effective concept observed was the use of enzyme formulations (as the mixture of endoglucanase, xylanase and pectinase), in comparison with isolated enzymes [8].

Enzyme formulations have been widely used in continuous liquid AD systems [9], however, to the best of our knowledge, there are no reports on their use in LBRs (i.e., batch conditions). In full-scale continuous stirred-tank reactors (CSTR) operated in continuous feeding mode, the formulation is added with the substrate in doses ranging

from 0.1 to 1 kg ton⁻¹TS [10]. The main challenge for using an enzyme formulation in LBRs lies in determining the best strategy to adopt: using it during the pre-treatment of substrates; adding it with the substrates at the beginning of the batch; or incorporating it into the reactor at different stages of the process.

Thus, the objective of the present study was to investigate different strategies for using enzyme formulations in the co-digestion of cattle dung and wheat straw in LBRs to avoid inhibition of the process by the accumulation of intermediate products, maximize biomass degradation and increase the methane yield.

2. Material and methods

2.1 Materials

The physico-chemical characteristics of the substrates (cattle dung and wheat straw), the solid digestate and stabilized leachate used for reactor inoculation, as well as the amounts used in the experiments are described in Table 1. The leachate and the digestate were collected from a full-scale LBR fed with manures and catch crop silage. The enzyme formulation used (UltraPract® HP by BIOPRACT) contained cellulase, xylanase, beta-glucanase and protease enzymatic activities.

Table 1 – Characterization and experimental conditions in terms of reactor feed.

		Wheat straw	Cattle dung	Solid digestate	Leachate
	pH	-	6.52 ± 0.02	-	7.95 ± 0.02
Alkalinity	(gCaCO ₃ L ⁻¹)	-	1.64 ± 0.23	-	11.01 ± 0.09
TS	(% FM)	89.94 ± 0.14	15.03 ± 0.19	19.46 ± 0.23	2.66 ± 0.01
VS	(% FM)	86.05 ± 0.08	12.96 ± 0.16	13.47 ± 0.36	1.41 ± 0.01
COD	(g L ⁻¹)	-	-	-	27.10 ± 1.29
TKN	(g Kg ⁻¹)	6.03 ± 0.11	6.12 ± 0.36	9.33 ± 0.42	2.52 ± 0.03
NH ₄ ⁺	(g Kg ⁻¹)	-	0.23 ± 0.04	1.72 ± 0.01	1.39 ± 0.02
HAc	(g L ⁻¹)	-	-	-	0.98 ± 0.08

Wet weight	(g)	332.02	916.13	514.46	1631.22
TS	(g)	289.62	137.72	100.11	43.41
VS	(g)	277.11	118.76	69.28	28.89

FM: fresh matter.

HAc: acetic acid

2.1 Reactors setup and co-digestion

Eight LBR reactors, with setups similar to that used by Thomas et al. [11], were used to conduct parallel experiments in duplicates. Briefly, the 7-L reactors were equipped with double jackets for heating (37°C) through thermoregulated baths, peristaltic pumps with timers for total recirculation of the leachate every two and a half hours (flow rate of 1 L min⁻¹), and Ritter gas flow meters for continuous volumetric monitoring of biogas production.

All reactors were fed with a mixture of 30% cattle dung and 70% wheat straw (on a volatile solid basis). After manually mixing the substrates and solid digestate, the solid phase of the reactor was compacted with a total weight of eight kilograms to simulate the conditions in a full-scale LBR. The leachate was then added and recirculated for fifteen minutes to saturate the solid phase with water. Before sealing the reactor, it was flushed for five minutes with nitrogen gas to ensure the absence of oxygen.

2.2 Strategies for enzyme formulation treatment

The enzyme formulation dose was set at 5 g per kg of VS in the substrate mixture. The first strategy (Prt) involved a conventional pre-treatment of the substrate mixture before its introduction into an LBR. The formulation was diluted in water and added to the mixture, maintaining a 1:2 ratio between straw and the formulation solution (wet weight). The pre-treatment was conducted at 37°C for 60 hours. To account for potential

biogas production during pre-treatment, it was carried out in the reactors, allowing for the use of the Ritter gas counter and gas sample bag for qualitative and quantitative analysis of the gas released.

The second strategy (Full) consisted of adding the full dose of the enzyme formulation at the start of the digestion. The third strategy (Half) involved adding half the dose of the enzyme formulation at the start and the other half on day 30. To ensure similar conditions to the pre-treatment scenario, all substrate mixtures received the same volume of water. In the Half condition, the enzyme formulation addition on day 30 was performed directly into the leachate through injection into the recirculation system. After this addition, the leachate in all reactors was continuously recirculated for one hour. Finally, a control reactor was operated without the addition of enzyme formulation (Ctrl).

2.3 Analysis

Physical-chemical analyses were performed at the Bio2E platform (Bio2E, 2018). The substrate, leachate and digestate were characterized in terms of TS, VS, TKN, NH_4^+ , pH, alkalinity and volatile fatty acids (VFAs). For characterization in terms of TS and VS, the samples were subjected to temperatures of 105 °C for 24 h and 550 °C for 2 h, respectively. Dunod's methodology [12] was used for TKN and NH_4^+ analyses. The analysis of pH, alkalinity [21] and VFAs were also used to monitor the digestion process. Water extraction was used when necessary (for this, 30 g of fresh solid sample and 150 mL of deionized water were homogenized in a mixing table during one hour at 30 °C and 250 RPM [14]. After, the mixture was centrifuged to separate and store the liquid phase to perform the analysis. VFAs concentrations were measured with a Perkin Clarus 580 gas chromatograph and the biogas composition was monitored with a Perkin Clarus 480

gas chromatograph, detecting CO₂, O₂, H₂S, N₂ and CH₄. To evaluate the impact of pretreatment and co-treatments on the substrates, the fiber composition of the substrate mixture was analyzed according to Van Soest [15] before and after treatment with the enzyme formulation. The genomic DNA of the microbial community from the samples, collected from liquid and solid phases after 90 days of reactor operation (without replication), was extracted using the FastDNA™ SPIN Kit for Soil DNA Extraction (MP Biomedicals) following the manufacturer's introductions. The sequencing and basic bioinformatics service for taxonomic identification were carried out by the company *NGS Soluções Genômicas* (Piracicaba, São Paulo – Brazil). For amplification of 16S rRNA gene (V3+V4 region), the primer sets 341F/785R [16] for the Bacteria Domain and Ar915aF/Ar1386R [17] for the Archaea were used. Sequencing libraries were generated using the DNA library UltraNBNext ®Kit for Illumina (New England Biolabs, Ipswich, USA), according with the manufacturer's recommendations. The sequencing data was processed using the DADA2, and taxonomies were assigned to each amplicon sequencing variant (ASV) through an implementation of the DADA2 program [18]. The SILVA database [19] served as the reference for taxonomic classification. Sequence data can be accessed in the National Center for Biotechnology Information (NCBI) database under the accession number PRJNA1167739 (SRX 26305809 - SRX 26305818, SRX 26305820, SRX 26305821, SRX 26305827 and SRX 26305828).

3. Results and discussion

3.1 Impact of enzymatic pretreatment on the substrate

A primary enzymatic treatment strategy involved using the enzyme formulation as a pretreatment step before the anaerobic digestion process. Table 2 presents the physico-chemical characteristics of the initial solid phase (substrate, mixture of substrate

and digestate) with enzymatic pretreatment (condition Prt) and without enzymatic pretreatment (conditions Ctrl, Full, and Half). Pretreatment altered the composition of the lignocellulose biomass, possibly increasing the susceptibility of the substrates to hydrolysis. It was verified an increase of 8.9% in the soluble fraction and a reduction of 10.6% in hemicellulose after pretreatment. An increase of 68.71% in the concentration of dissolved Chemical Oxygen Demand (COD_D) was observed under the condition with pre-treatment. Furthermore, the pre-treatment resulted in a slight decrease in the solid content.

Table 2 – Characterization of the initial solid phase in the reactors without pre-treatment (Ctrl, Full and Half) and with enzymatic pre-treatment (Prt).

Parameter	Without pre-treatment	With pre-treatment
Substrate		
Soluble fraction (%)	18.8 ± 0.2	20.5 ± 0.2
Hemicellulose-like (%)	28.5 ± 0.8	25.5 ± 0.3
Cellulose (%)	36.2 ± 0.6	39.0 ± 0.7
Lignin (%)	16.5 ± 0.8	15.0 ± 0.9
Substrate and digestate mixture		
pH	6.3 ± 0.2	5.9 ± 0.1
TS (% FM)	23.1 ± 0.4	22.0 ± 0.4
VS (% FM)	26.5 ± 0.8	24.6 ± 0.8
COD _D (g Kg ⁻¹)	11.4 ± 1.6	19.2 ± 2.1
CHO _D (g Kg ⁻¹)	0.07 ± 0.01	0.05 ± 0.01
HAc (g Kg ⁻¹)	1.5 ± 0.1	46 ± 2
HPr (g Kg ⁻¹)	< DL	13.0 ± 0.9
HBu (g Kg ⁻¹)	< DL	10.9 ± 0.7

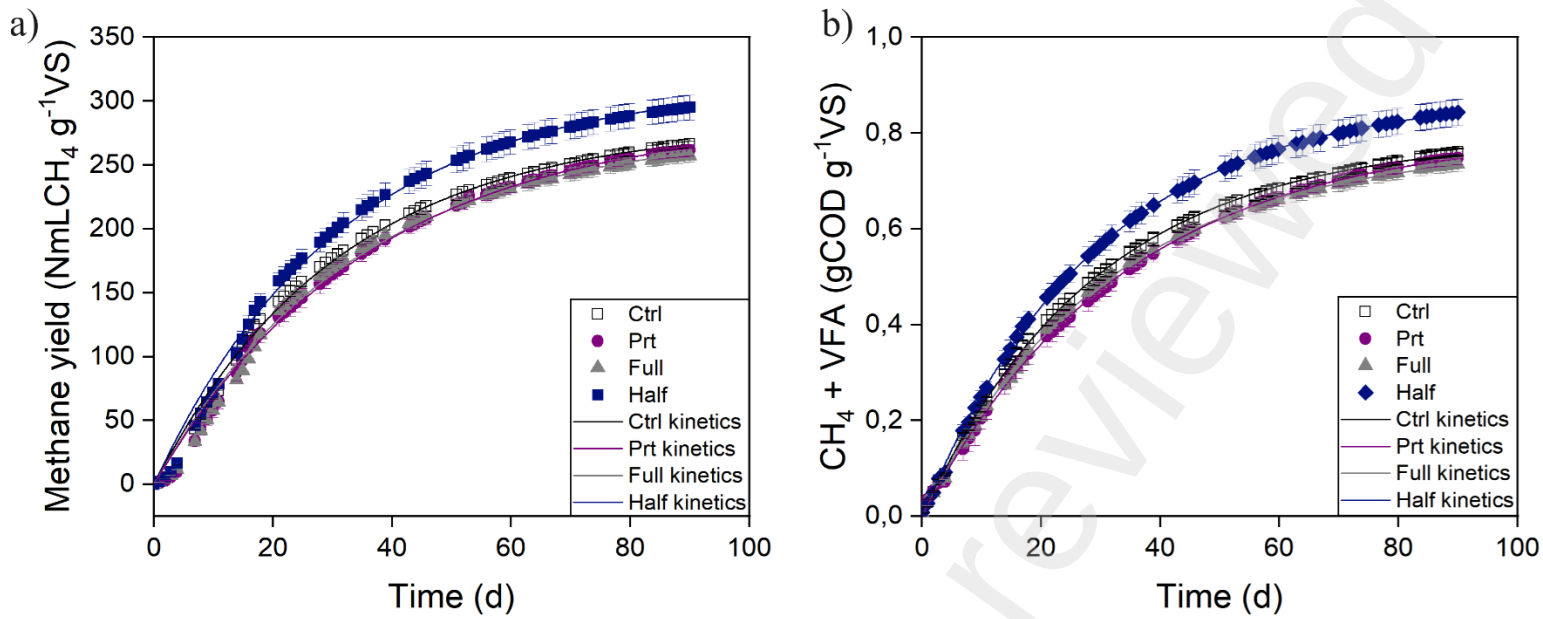
During the 60-hour pretreatment, a small production of CO₂ and CH₄ was observed, with methane production corresponding to a methane yield of 1.9 ± 0.1 NmL CH₄ g⁻¹VS. This occurred because the anaerobic digestion process begun during the pretreatment step due to the mesophilic temperature used and the presence of microorganisms in the cattle dung. Indeed, several studies investigated the use of cattle manure as inoculum for this type of process [20]. However, the production observed in this step was very small compared to the biochemical methane production (BMP) of the

substrates. Indeed classical values of BMP are between 200 and 230 NmL CH₄ g⁻¹VS for cattle manure [11,21], and from 237 to 303 NmL CH₄ g⁻¹VS for wheat straw [22,23].

Comparing the initial solid composition (substrate + digestate) between pretreated and other conditions, a production of acetic acid (HAc), propionic acid (HPr) and butyric acid (HBu) was observed during pretreatment. This accumulation of volatile fatty acids (VFA) was associated with a pH decrease from 6.3 ± 0.2 to 5.9 ± 0.1 in the initial solid mixture.

3.2 Anaerobic digestion performance

The use of enzymes, as well as the different strategies for their use (pre-treatment or addition), significantly affected the reaction dynamics in the LBR reactor, impacting both the leachate composition and the biogas production. Figure 1 presents the characteristics of the leachate during the digestion process in the LBR for the four tested conditions. Although pre-treatment reduced the pH of the initial solid phase, the impact on the pH of the leachate at time zero was slight and did not significantly affect the process startup. While the initial pH in Prt was 7.93 ± 0.04 , in the other reactors the values ranged from 8.11 ± 0.05 to 8.17 ± 0.02 . A slight reduction in initial alkalinity was observed between conditions Ctrl (9.29 ± 0.03 gCaCO₃ L⁻¹) and Prt (9.09 ± 0.01 gCaCO₃ L⁻¹) due to the production of VFAs during pre-treatment with the enzyme formulation (Figure 1b).



	Ctrl	Prt*	Full	Half
Methane yield after 60 days (NmL g ⁻¹ VS)	240 ± 4	233 ± 4	231 ± 5	268 ± 10
Methane yield after 90 days (NmL g ⁻¹ VS)	266 ± 2	262 ± 4	257 ± 5	295 ± 10
Methane yield with first-order adjustment (NmL g ⁻¹ VS)	282 ± 2	287 ± 3	278 ± 3	314 ± 3
k (d ⁻¹)	0.032 ± 0.001	0.028 ± 0.001	0.030 ± 0.001	0.032 ± 0.001
Hydrolysis yield with first-order adjustment (gCOD g ⁻¹ VS)	0.791 ± 0.002	0.803 ± 0.002	0.773 ± 0.002	0.881 ± 0.002
k (d ⁻¹)	0.034 ± 0.001	0.030 ± 0.001	0.033 ± 0.001	0.034 ± 0.001

Figure 1 – Accumulated methane yield and its first-order kinetics, (b) Hydrolysis yield (sum of VFAs and methane production) and its first-order kinetics and c) Methane yield with 60 and 90 days of digestion, and methane and hydrolysis yield with first order adjustment and its kinetics. *without the 1.9 ± 0.1 NmL g⁻¹VS produced during the pretreatment

For all conditions, pH decreased drastically during the first four days but remained close to neutrality (Figure 1a). After this period, an increase in pH was observed until around day 15, followed by stabilization. Regarding VFAs accumulation, conditions Ctrl, Prt and Half reached the VFAs peak after three days (19.1 ± 0.2 gCOD, 21.0 ± 1.4 gCOD and 19.7 ± 1.3 gCOD, respectively), whereas in Full condition (addition of the full dose at the startup of the process), the VFAs concentration increased and accumulation

continued until the seventh day (22.2 ± 1.9 gCOD). This delay could indicate some inhibition of methanogens, suggesting that the full dose of enzyme formulation was higher than the process could support at startup.

Among the VFAs present at the beginning and accumulated during anaerobic digestion, HAc and HPr were predominant (Figure 1d and 1e). Although in lower concentrations, HBu was also detected (Figure 1f). Based on high production of HBu, it is possible to infer that some H_2 , not measured, was released. For all the reactors, HAc was only detected until day 43, HPr until day 21 and HBu until day 9. After adding half dose of the enzyme formulation on day 30, a drop in the pH and alkalinity of the leachate was observed. However, no accumulation of VFAs was verified (Figure 1c). Despite the probable production of VFAs, indicated by the drop in pH and alkalinity, the consumption occurred at the same rate as production. This likely happened due to the stability and the microbial growth that occurred during this period.

Riggio et al. [4] described that the operation of LBRs can be divided in three periods. Period I is the acidification phase, which can last up to around 15 days. During this period, acidification occurs due to hydrolysis, acidogenesis, acetogenesis and the lag phase of methanogenesis. Following this period, methanogenesis becomes established (period II). Then, between about days 35 and 90, the degradation completion phase is observed (period III). However, the duration of each phase was based on LBRs fed with spent cow bedding and cereal waste. In the case of the fresh cattle dung used in this present study, the duration of the first period was reduced due to its alkalinity.

After seven days of operation, the methane content in biogas was around 60% in all the reactors, indicating the establishment of methanogenesis. The methane content in

the biogas during periods II and III was around 55% under all conditions, with a general average of $55.4 \pm 0.1\%$.

Despite the fact that an increase in accumulated methane yield (*MY*) without adjustment was observed between days 60 and 90 of digestion, this increase was slight, with a general average of $11.1 \pm 0.1\%$ (Figure 2). Among the strategies tested, an increase in *MY* relative to Ctrl was observed only in the Half condition. Considering the standard digestion time used in full-scale LBRs (60 days), the increase was 11.7%.

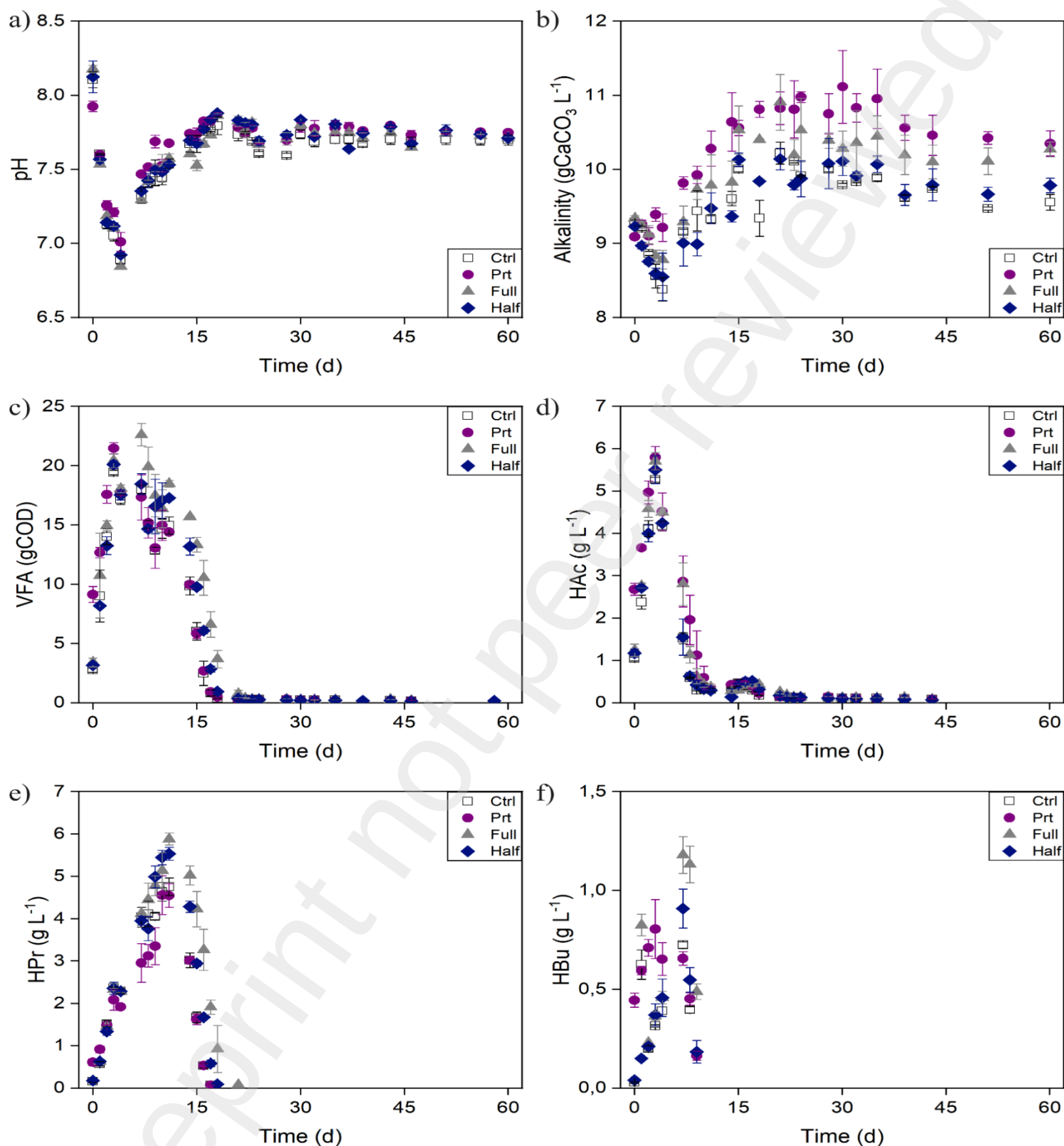


Figure 2 – Characterization of the leachate during the digestion process on LBR using enzyme formulation in terms of pH (a), alkalinity (b), total VFAs (c), acetic acid (d), propionic acid (e) and butyric acid (f).

The adjustment using a first-order model was performed for the methane yield (*P*) and the hydrolysis yield (*HY*). *HY* was calculated as the sum of CH₄ and VFAs production expressed as equivalent COD. The Half condition promoted an increase of 11.3% in *P* and 11.4% in *HY* compared to the control after 90 days of digestion. However, no increase in the kinetics was observed for this condition. For the other strategies (Prt and Full), a decrease in both kinetics of *P* and *HY* was observed compared to the control. It is possible that the pre-treatment and the addition of the full dose of enzyme formulation on day zero resulted in slight inhibition of microbial activity. Moreover, in the Full dose addition condition, not only methanogenesis but also hydrolysis was inhibited. Thus, in the specific case of batch LBR, it is important not to overdose the enzymes relative to microbial growth. Excessive enzyme dose initially introduces a surplus of readily accessible sugars, leading increased VFAs production and reduced rate of methanogenesis. The most effective use of enzymes involves adding them at the startup and several days (30 days for example) after, allowing sufficient microbial growth.

Although literature results indicate an increase in methane production of 61 to 91% with enzyme formulations for pretreating agricultural waste or directly adding them to reactors [6,24], these findings typically stem from BMP tests under controlled conditions (low organic load) optimized for maximizing methane production and enzyme activity. However, Bhatnagar et al. [24] also noted methanogenic inhibition in a BMP test when the concentration of an enzyme formulation (composed of cellulase, xylanase and beta-glucanase) was increased during the treatment of grass silage and chicken litter for the biogas production. It should be noted that the positive effect observed when adding half doses at different times (Half condition) aligns with the manufacturer's expectations (increase in biogas yield of 10 to 20%), although the enzyme formulation dose preconized

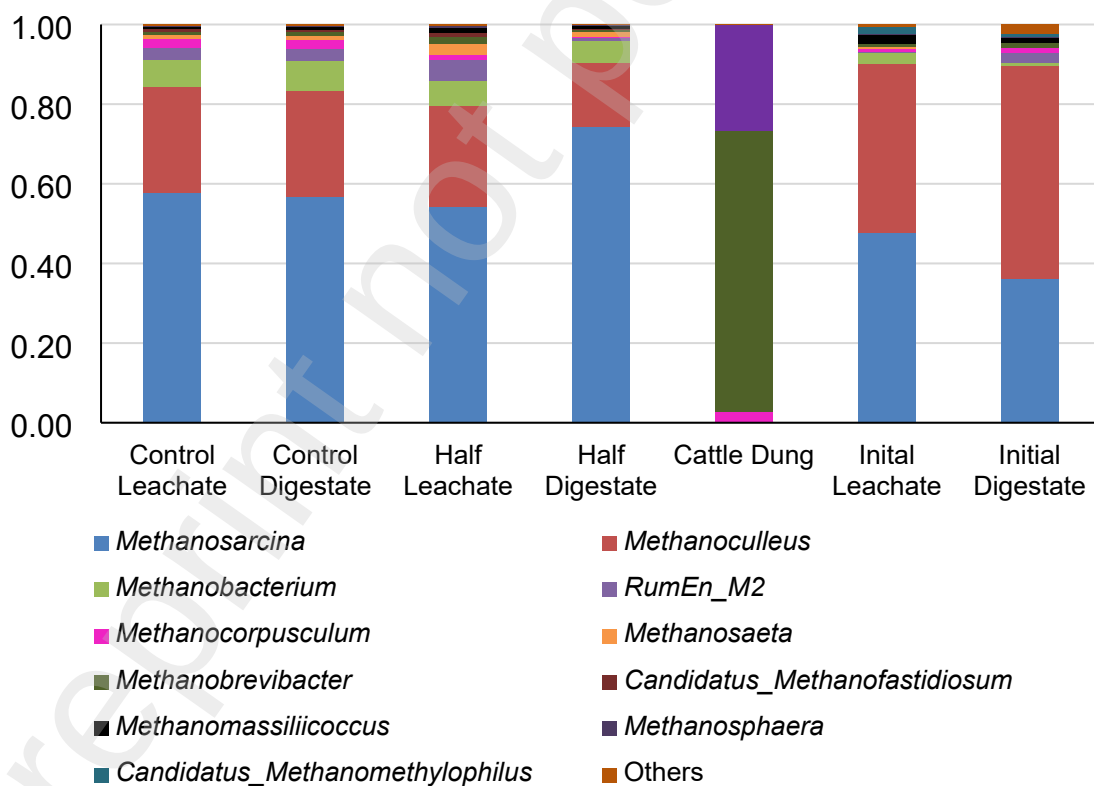
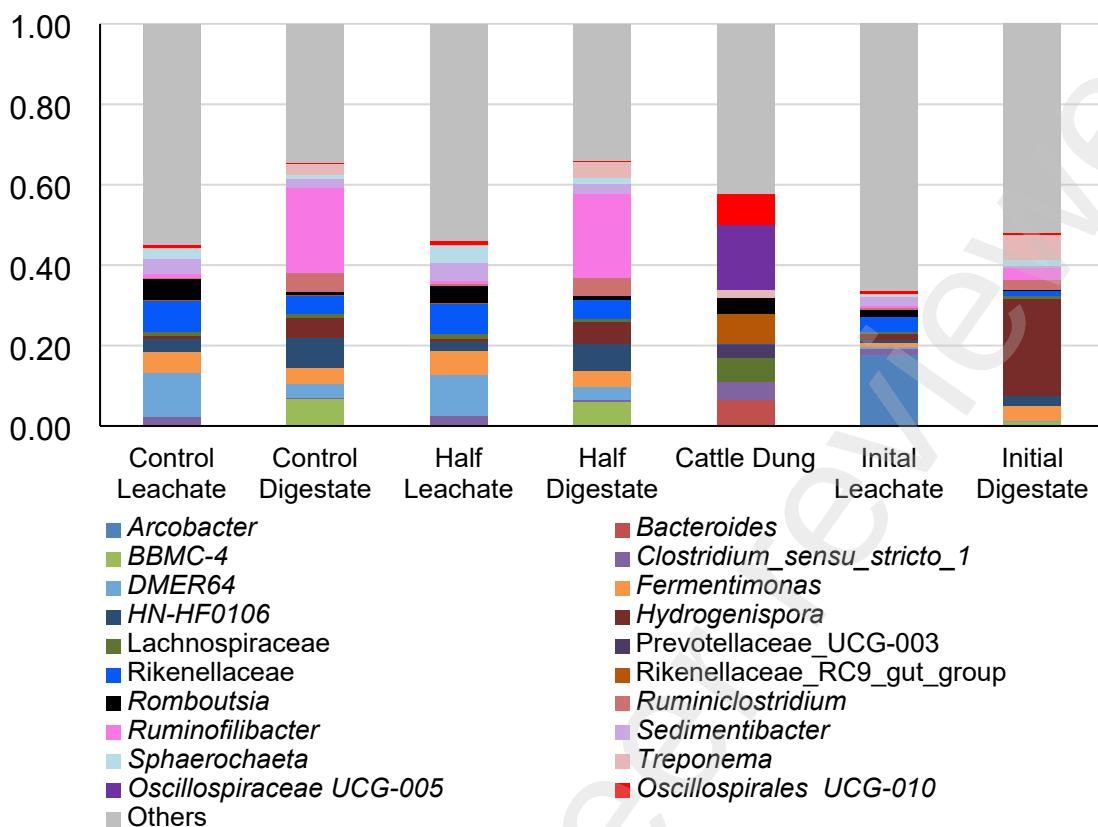
in full-scale plants (0.12 g Kg⁻¹VS) with a continuous reactor is lower than that used in the present study.

3.3 Microbial analysis

To assess the impact of using half the dose of the enzyme formulation on the microbial community, the final samples of the leachate and solid digestate from the Ctrl condition and Half condition were analyzed. Both solid (digestate) and liquid (leachate) phase were examined, as differences in the microbial community profile are expected between these phases within the same reactor. In the solid phase hydrolysis, acidogenesis and acetogenesis activity (phases associated with bacteria) predominate, while in the liquid phase, methanogenesis is more predominant. For the bacterial domain (Figure 3a), no significant changes in the relative abundance (> 3%) of the identified genera were observed. The genera with relative abundances above 5% in at least one of the analyzed leachate samples were *DMER64* (10.4% in Ctrl and 10.4% in Half), *Rikenellaceae* (7.6% in Ctrl and 7.4% in Half), *Fermentimonas* (5.2% in Ctrl and 5.9% in Half), and *Romboutsia* (5.07% in Ctrl and 4.38% in Half). The genus *DMER64* is functional microorganism involved in interspecies hydrogen transfer (IHT), contributing to the reduction of hydrogen pressure [25]. Additionally, Lee et al. [26] describes it as a butyrate beta-oxidizing bacterium that grows significantly under ammonia-stressed conditions, making it a potential syntrophic bacterium to establish direct interspecies electron transfer (DIET) with methanogens to accelerate methanogenic degradation of butyrate under ammonia stress. According to Li et al. [27], *DMER64* shows a positive correlation with *Methanosarcina*. Interestingly, codigestion led to a notable increase in relative abundance of this genus, rising from below 2% in initial leachate and digestate

samples. The influence of cattle dung as an inoculum is also highlighted, with a relative abundance of 7.51% for *Rikenellaceae_RC9_gut_group* in the residue. Rikenellaceae are potential syntrophic organisms capable of DIET during the degradation of volatile fatty acid (VFA) for methane production. These bacteria are often co-cultured with *Methanosarcina* species, demonstrating their relevance in DIET processes [28].

a)



315
316 Figure 3 – Relative abundance of main genera identified in leachate and digestate of Ctrl and Half and in
317 initial leachate, digestate and cattle dung for (a) bacteria (>3%) and for (b) Archaeas (>1%).
318

In the bacterial domain of the solid digestate, genera with a relative abundance higher than 5% in at least one sample included *Ruminofilibacter* (18.8% in Ctrl and 20.81% in Half), HN-HF0106 (7.47% in Ctrl and of the solid digestate 6.7% in Half), BBMC-4 (6.7% in Ctrl and 6.1% in Half), and *Hydrogenispora* (4.7% in Ctrl and 5.4% in Half). Among these genera, the identification of *Hydrogenispora* stands out in both initial leachate and digestate, with relative abundances of 1.4% and 14.7%, respectively. The predominance of *Ruminofilibacter*, a degrader of lignocellulosic materials [3], is significant, and it is also commonly found in manures [29].

In the archaea domain, no effect of using half the dose of the enzyme formulation was observed in the final leachate (Figure 3b). The relative abundances of the predominant genera were quite similar: *Methanosarcina*, (57.7% in Ctrl leachate and 54.2% in Half leachate) *Methanoculleus* (26.8% in Ctrl leachate and 25.3% in Half leachate). In the solid digestates, however, a marked increase of *Methanosarcina* genus was observed with the use of half the dose of the enzyme formulation, rising from 56.7% in Ctrl digestate to 74.3% in Half digestate. In contrast, the relative abundance of *Methanoculleus* decreased with the enzyme formulation addition (26.7% in Ctrl and 16.12% in Half). In the initial samples of solid digestate, both genera were predominant (42.3% for *Methanoculleus* and 53.7% for *Methanosarcina*) and cow dung (47.8% for *Methanoculleus* and 36.0% for *Methanosarcina*). Both genera are well-documented as predominant in archaea community of leach-bed reactors [30-32]. *Methanoculleus* is involved in hydrogen utilization and demonstrates strong tolerance to ammonia, explaining its dominance high-ammonia environment [33] like initial cow dung. On the other hand, *Methanosarcina* can produce methane by consuming different intermediates, such as HAc, hydrogen, and methyl-containing compounds [34]. Its dominance may be

attributed to its syntrophic relationship with the DMER64, a member of the Rikenellaceae family.

4. Conclusion

Only the addition of the half dose of formulation on day zero plus the other half dose on day 30 resulted in an increase of methane yield (11.7%), in this condition, *Methanosarcina* was dominant in the solid phase (74.3%). The main bacteria involved in hydrolysis, acidogenesis, and acetogenesis in the solid phase was *Ruminofilibacter* (21%). Under other conditions, an inhibition of the process was observed. In the condition involving substrate pretreatment, the changes in the composition of lignocellulosic biomass due to pretreatment increased the substrates' susceptibility to hydrolysis, resulting in higher VFA production during the startup phase of the process and inhibition of methanogenesis. In the condition with full-dose addition, for instance, both methanogenesis and hydrolysis were inhibited. Thus, in the specific case of batch LBR, it is important not to overdose the enzymes relative to microbial growth. A high dose of enzymes at initial time will lead to a high amount of readily accessible sugars that will be converted into VFA and the rate limiting stage for hydrolysis and methanogenesis. The most beneficial use of enzymes consists in their addition several days (30 days for example) after the start-up, when microorganisms have grown.

Acknowledgements

The authors are grateful for the financial support of the State of São Paulo Research Foundation (FAPESP) for the Research Internship Abroad (process 2022/11092-2) and

INRAE Bio2E Facility (Bio2E, INRAE, 2018). Environmental Biotechnology and Biorefinery Facility (<https://doi.org/10.15454/1.557234103446854E12>) where all experiments were conducted. Biopract ABT GmbH is acknowledged for providing enzyme formulation.

References

- [1] Kuthiala, T., Thakur, K., Sharma, D., Singh, G., Khatri, M., Arya, S. K., 2022. The eco-friendly approach of formulation enzyme in agricultural waste treatment: A comprehensive review. *Int. J. Biol. Macromol.* 209, 1956-1974.
- [2] Atinkut, H. B., Yan, T., Arega, Y., Raza, M. H., 2020. Farmers' willingness-to-pay for eco-friendly agricultural waste management in Ethiopia: A contingent valuation. *J. Clean. Prod.* 261, 121211.
- [3] Shen, J., Zheng, Q., Zhang, R., Chen, C., Liu, G., 2019. Co-pretreatment of wheat straw by potassium hydroxide and calcium hydroxide: Methane production, economics, and energy potential analysis. *J. Environ. Manage.* 236, 720-726.
- [4] Riggio, S., Torrijos, M., Vives, G., Esposito, G., van Hullebusch, E. D., Steyer, J. P., Escudié, R., 2017. Leachate flush strategies for managing volatile fatty acids accumulation in leach-bed reactors. *Bioresour. Technol.* 232, 93-102.
- [5] Wang, Z., Hu, Y., Wang, S., Wu, G., Zhan, X., 2023. A critical review on dry anaerobic digestion of organic waste: Characteristics, operational conditions, and improvement strategies. *Renew. Sustain. Energy Rev.* 176, 113208.
- [6] Anacleto, T. M., Oliveira, H. R., Diniz, V. L., de Oliveira, V. P., Abreu, F., Enrich-Prast, A., 2022. Boosting manure biogas production with the application of pretreatments: A meta-analysis. *J. Clean. Prod.* 362, 132292.

- [7] Afolalu, S. A., Salawu, E. Y., Ogedengbe, T. S., Joseph, O. O., Okwilagwe, O., Emetere, M. E., ... Akinlabi, S. A., 2021. Bio-agro waste valorization and its sustainability in the industry: a review. In IOP conference series: materials science and engineering (Vol. 1107, No. 1, p. 012140). IOP Publishing.
- [8] Hao, X., Wen, P., Wang, J., Wang, J., You, J., Zhang, J., 2020. Production of xylooligosaccharides and monosaccharides from hydrogen peroxide-acetic acid-pretreated poplar by two-step enzymatic hydrolysis. *Bioresour. Technol.* 297, 122349.
- [9] Brémond, U., de Buyer, R., Steyer, J. P., Bernet, N., & Carrere, H., 2018. Biological pretreatments of biomass for improving biogas production: an overview from lab scale to full-scale. *Renewable and Sustainable Energy Reviews*, 90, 583-604.
- [10] Leca, E., Zennaro, B., Hamelin, J., Carrère, H., & Sambusiti, C., 2023. Use of additives to improve collective biogas plant performances: A comprehensive review. *Biotechnology Advances*, 108129.
- [11] Thomas, H. L., Seira, J., Escudié, R., Carrère, H., 2018. Lime pretreatment of miscanthus: impact on BMP and batch dry co-digestion with cattle manure. *Molecules* 23(7), 1608.
- [12] Rodier, Dunod. *L'analyse de l'eau*. 1975, vol. 1, p : 123-126.
- [13] Hill, D. T., Jenkins, S. R., 1989. Measuring alkalinity accurately in aqueous systems containing high organic acid concentrations. *Trans. ASAE*, 32(6), 2175-2178.
- [14] Van Vlierberghe, C., Escudié, R., Bernet, N., Frederic, S., Carrère, H., 2021. Long term alkaline storage and pretreatment process of cover crops for anaerobic digestion. *Bioresource Technology*, 330, 124986.

- 413 [15] Van Soest, P. J., Wine, R. H., 1967. Use of detergent in the analysis of fibrous
414 feeds IV. Determination of plant cell wall constituents. *J. Asso. Anal. Chem.* 50(1), 50-
415 55.
- 416 [16] Klindworth A., Pruesse E., Schweer T., Peplles J., Quast C., et al., 2013a.
417 Evaluation of general 16S ribosomal RNA gene PCR primers for classical and
418 next-generation sequencing-based diversity studies. *Nucleic Acids Res* 41(1).
- 419 [17] Kittelmann, S., Seedorf, H., Walters, W. A., Clemente, J. C., Knight, R.,
420 Gordon, J. I., & Janssen, P. H., 2013b. Simultaneous amplicon sequencing to explore
421 co-occurrence patterns of bacterial, archaeal and eukaryotic microorganisms in rumen
422 microbial communities. *PloS one*, 8(2):e47879.
- 423 [18] Wang Q., Garrity G.M., Tiedje J.M., Cole J.R., 2007. Naive bayesian classifier
424 for rapid assignment of rRNA sequences into the new bacterial taxonomy. *Applied and*
425 *environmental microbiology*, 73:5261–7.
- 426 [19] Glöckner F.O., Yilmaz P., Quast C., Gerken J., Beccati A., Ciuprina A., et al.,
427 2017. 25 years of serving the community with ribosomal RNA gene reference databases
428 and tools. *Journal of biotechnology*, 261:169–76.
- 429 [20] Rodrigues, C. V., Camargo, F. P., Lourenço, V. A., Sakamoto, I. K.,
430 Maintinguer, S. I., Silva, E. L., Varesche, M. B. A., 2024. Towards a circular
431 bioeconomy to produce methane by co-digestion of coffee and brewery waste using a
432 mixture of anaerobic granular sludge and cattle manure as inoculum. *Chemosphere*,
433 357.
- 434 [21] Kumar, P., Prajapati, S. K., Malik, A., Vijay, V. K., 2017. Cultivation of native
435 algal consortium in semi-continuous pilot scale raceway pond for greywater treatment

coupled with potential methane production. Journal of environmental chemical engineering, 5(6), 5581-5587.

[22] Peng, X., Börner, R. A., Nges, I. A., Liu, J., 2014. Impact of bioaugmentation on biochemical methane potential for wheat straw with addition of *Clostridium cellulolyticum*. Bioresource technology, 152, 567-571.

[23] Victorin, M., Davidsson, Å., Wallberg, O., 2020. Characterization of mechanically pretreated wheat straw for biogas production. Bioenergy research, 13(3), 833-844.

[24] Bhatnagar, N., Ryan, D., Murphy, R., Enright, A. M., 2019. Effect of co-digestion ratio and enzyme treatment on biogas production from grass silage and chicken litter. Waste Biomass Valorization 10, 3271-3277.

[25] Xie, Z., Huang, S., Wan, Y., Deng, F., Cao, Q., Liu, X., & Li, D. (2023). Power to biogas upgrading: Effects of different H₂/CO₂ ratios on products and microbial communities in anaerobic fermentation system. Science of The Total Environment, 865, 161305.

[26] Lee, J., Koo, T., Yulisa, A., & Hwang, S., 2019. Magnetite as an enhancer in methanogenic degradation of volatile fatty acids under ammonia-stressed condition. Journal of environmental management, 241, 418-426.
<https://doi.org/10.1016/j.jenvman.2019.04.038>

[27] Li, Y., Chen, Z., Ye, Z. L., & Lyu, R (2023). Determination Of The Effects of Salt and Oil on Substance Transformation, Microbial Community, and Transcriptional Activity During Anaerobic Digestion of Food Waste Through Integrated 16s Rrna Sequencing Technology and Metatranscriptomic Analysis.

- [28] Guan, R., Yuan, H., Zhang, L., Zuo, X., & Li, X., 2021. Combined pretreatment using CaO and liquid fraction of digestate of rice straw: Anaerobic digestion performance and electron transfer. *Chinese Journal of Chemical Engineering*, 36, 223-232.
- [29] Jang, H. M., Brady, J., & Kan, E., 2021. Succession of microbial community in anaerobic digestion of dairy manure induced by manure-derived biochar. *Environmental Engineering Research*, 26(1).
- [30] Li, Y., Wang, Z., Li, T., Jiang, S., Sun, Z., Jiang, H., ... & Xu, Q., 2020. Changes in microbial community and methanogenesis during high-solid anaerobic digestion of ensiled corn stover. *Journal of Cleaner Production*, 242, 118479.
- [31] Yang, J., Wang, D., Luo, Z., Zeng, W., & Huang, H., 2020. The role of reflux time in a leach bed reactor coupled with a methanogenic reactor for anaerobic digestion of pig manure: reactor performance and microbial community. *Journal of Cleaner Production*, 242, 118367.
- [32] Xing, T., Kong, X., Dong, P., Zhen, F., & Sun, Y., 2020. Leachate recirculation effects on solid-state anaerobic digestion of Pennisetum hybrid and microbial community analysis. *Journal of Chemical Technology & Biotechnology*, 95(4), 1216-1224.
- [33] Xiao, Y., Yang, H., Zheng, D., Liu, Y., Zhao, C., & Deng, L., 2021. Granular activated carbon alleviates the combined stress of ammonia and adverse temperature conditions during dry anaerobic digestion of swine manure. *Renewable Energy*, 169, 451-460.
- [34] Tang, F., Tian, J., Zhu, N., Lin, Y., Zheng, H., Xu, Z., & Liu, W., 2022. Dry anaerobic digestion of ammoniated straw: Performance and microbial characteristics. *Bioresource Technology*, 351, 126952.

Preprint not peer reviewed